11.8 Evaluating Numeric Expressions – Part 1

Lesson Introduction

Benchmarks	MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value. MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.	Learning Target	• I can evaluate numeric expressions with up to four operations.

Benchmark Coherence	Building On	Working Toward
	MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations. MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.	MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Essential Questions	• How can I use order of operations to evaluate numeric expression? • How are numeric expressions with multiple grouping symbols evaluated?
Lesson Narrative	In this lesson, students extend their knowledge of the order of operations to include what they have learned about exponents, the use of the absolute value symbol, and multiple grouping symbols. Students begin by getting a refresher on the process of the order of operations that they learned in 5th grade and receive formal instruction on using the order of operations with exponents and grouping with a focus on absolute value notation. Students then get the opportunity to practice applying what they have learned by evaluating numeric expressions.

Component Summary				
	11.8.1 Warm-Up (~5 min)	Math symbols activity *(SE p. 200)*	11.8.4 Exploration (10-15 min)	Collaborative exploration activity on grouping symbols *(SE p. 203)*
	11.8.2 Guided Instruction (10-15 min)	Instruction on the order of operations with exponents, absolute value, and grouping symbols *(SE p. 200)*	11.8.5 Your Turn! (5-10 min)	Independent practice on evaluating expressions *(SE p. 204)*
	11.8.3 Collaboration (10 min)	Collaborative practice with evaluating expressions *(SE p. 203)*	11.8.6 Wrap-Up (~5 min)	Students reflect on their learning *(SE p. 205)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms</u>: • Nested grouping symbols • Absolute value notation <u>Additional Terms</u>: • Order of operations	• (Optional) Colored markers or highlighters • (Optional) Calculators • (Optional) Video	• Consider grouping for 11.8.3 Collaboration. • The 11.8.1 Warm-Up video is optional.

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may confuse the order of operations by doing all multiplication before division instead of from left to right in order. • Students may forget to complete the calculation inside the absolute value before writing its actual absolute value. • Students may forget to write the positive value when working with absolute values. • Students may misunderstand the hierarchy of operations.	• Students may have trouble with nested grouping symbols and identifying which numbers/operations are in the inner grouping vs the outer grouping. Consider using colored highlighters or other visual markers to help students see the process unfold. • Often, there is more than one way to evaluate a numeric expression by applying the order of operations, properties of equality, and laws of exponents. As you work the problems, consider alternate correct ways to evaluate each expression to be prepared for alternate correct suggestions from students. • The order of operations is often taught using a mnemonic device as a hard and fast rule. The lesson does not contain any mnemonic devices but instead relies upon both the order of operations and the properties of equality, such as commutative and associative properties. Encourage students' flexible thinking by demonstrating various ways a numeric expression can be evaluated. • If time is an issue, then consider assigning 11.8.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect In 11.8.2 Guided Instruction, students learn about evaluating expressions when nested grouping symbols are present. To help students understand this, consider demonstrating thinking out loud and prompting students to reflect and respond.	**MA.K12.MTR.4.1: Discussion and Reflection** During 11.8.2 Guided Instruction, students have the opportunity to analyze the mathematical thinking of two students. Discussing the two different methods with their partner helps students construct a justification that reflects the two correct methods. In 11.8.4 Exploration, students work with their partner to explore similar numeric expressions where they communicate the order of operations, properties of equality, and laws of exponents.
	11.8.4 Exploration have students working together in partner pairs. Consider pairing students based on the data from the previous lesson. This can help struggling students learn from their peers while strengthening the other students' understanding of the concept. 11.8.6 Wrap-Up activity in this lesson can provide valuable data on whether or not a student understands the concepts in this lesson.	

Lesson Notes:

11.8.1 Warm-Up: Math Symbols

Component Overview: *Students will engage in an activity on math symbols where they list symbols they are familiar with. They will distinguish between ones they are familiar with and symbols they have seen but are unfamiliar with.*

11.8.1 Warm-Up: Math Symbols

In the 18th century, Robert Recorde decided to use a symbol for the idea of "is equal to" so he would not have to write the words over and over. As more mathematicians also began to use the symbol, the "=" became accepted as the standard symbol for equality. Math is full of symbols that shorten lengthy words and calculations. But just like symbols you use when communicating with others, like "LOL" in texting, once you learn their meaning, understanding the mathematics is not so challenging.

1. List math symbols that you are familiar with and what they represent.
   Answers will vary. X means multiply, - means subtract, + means add.

2. List other math symbols you have seen before, but you are unfamiliar with.
   Answers will vary. $\pm$, $\cup$, $\approx$

Consider providing students with several symbols to choose from. Students may not be able to recall symbols that they have seen but do not understand.

11.8.2 Guided Instruction: Applying the Order of Operations for Expressions Including Exponents

Component Overview: *Students will receive formal instruction on using the order of operations with exponents, absolute value, and nested grouping symbols. For this activity, consider leveraging the "I Do" and "We Do" components of the gradual release model. Facilitate the questions in a discussion form to help students connect to the concepts.*

11.8.2 Guided Instruction: Applying the Order of Operations for Expressions Including Exponents

Recall from 5th grade that an order of operations is used to evaluate numeric expressions with multiple operations.

Step 1: Perform operations within grouping symbols.
Step 2: Multiply and divide in order from left to right $(\rightarrow)$.
Step 3: Add and subtract in order from left to right $(\rightarrow)$.

Numerical expressions can also include exponents.

1. The expression $4 \times (-2)^3 - 8 \div (-3 + 1)$ can be rewritten by expanding the exponential expression as $4 \times (-2 \cdot -2 \cdot -2) - 8 \div (-3 + 1)$.

   a. According to the order of operations, which expressions should be evaluated first?
   The expressions in parentheses should be evaluated first.

   b. When $(-2)^3$ is rewritten using multiplication, the resulting expression is in parentheses. Complete the statement.

   According to the order of operations, $(-2)^3$ should be evaluated [○ after / ○ before] multiplying by 4.

   before

Students will spend more time evaluating expressions with exponents in Unit 13 Algebraic Expressions. The goal here is to make a connection to exponents as grouped repeated multiplication to help students understand why exponents should be evaluated before multiplication and division in a numeric expression.

11.8.2 Guided Instruction: Applying the Order of Operations for Expressions Including Exponents (continued)

c. Work with your partner to number each operation so that the order of operations is in the correct order.

1	Perform operations inside grouping
3	Perform multiplication and division in order from left to right. (→)
2	Evaluate expressions with exponents.
4	Perform addition and subtraction in order from left to right. (→)

2. Use order of operations to evaluate one of the expressions shown in the table below while your partner evaluates the other expression.

Expression A	Expression B
$(6+0.5)^2 \div 0.25$	$(6^2+0.5^2) \div 0.25$
$6.5^2 \div 0.25$ $= 42.25 \div 0.25$ $= 169$	$(36+.25) \div 0.25$ $= 36.25 \div 0.25$ $= 145$

3. How are the problems similar?
Answers will vary. The problems are similar by both having exponents and the final step being to divide by 0.25.

4. How are the problems different?
Answers will vary. The problems are different in that in expression A, you add the expressions inside the parenthesis before applying the exponent, whereas in expression B you apply the exponents to the expressions inside the parentheses before taking their sum.

For questions 2 and 3, consider having students discuss them with a partner and then share out their responses in whole group. The responses may be more thorough when provided orally from students than their writing of the explanation in the book.

11.8.2 Guided Instruction: Applying the Order of Operations for Expressions Including Exponents (continued)

5. Complete the statements about the expressions.

The two expressions are
- o equivalent
- o not equivalent

because they
- o have the same value.
- o do not have the same value.

For Expression A, first
- o add inside the parentheses, then apply the exponent to the sum.
- o apply the exponent to each addend, then add to find the sum.

For Expression B, first
- o add inside the parentheses, then apply the exponent to the sum.
- o apply the exponent to each addend, then add to find the sum.

For both expressions, addition was done before division

because
- o the addition is to the left of the division.
- o the addition is inside the grouping symbols.

6. Anna evaluated Expression A and Expression B. Her work is shown.

Expression A	Expression B
$\lvert -10\rvert \div 2 + 10^2$ $10 \div 2 + 10^2$ $10 \div 2 + 100$ $5 + 100$ 105	$-10 \div 2 + 10^2$ $-10 \div 2 + 100$ $-5 + 100$ 95

Students may need additional strategies for how to approach these types of questions. Consider showing students how to break down each sentence in this series of statements and how to look for correct matches in each one.

How can you determine if two numerical expressions are equivalent?

- *What would have happened if the 10 was positive in both?*
- *Would the answers have been different still?*
- *Why or why not?*

11.8.2 Guided Instruction: Applying the Order of Operations for Expressions Including Exponents (continued)

Anna's partner gave her a 👍 on both of her answers. Why did Anna arrive at two different values when evaluating these similar looking expressions?
In expression A, you have to find the absolute value of -10*, making the value a positive* 10*.*

Absolute value notation should be treated the same as grouping symbols when following the order of operations.

7. Zion says since multiplication is commutative and the order does not matter, he thinks he can evaluate the expression $[0.05 \times (3 \times 2)^4]^2$ by either multiplying 0.05×3 first or by multiplying 3×2 first.

   a. Discuss with your partner if Zion is correct when he stated that both would work.
   Answers will vary. The result of the discussion should be that Zion is incorrect. There is an exponent that has to be applied first in this case, so you must multiply 3 *by* 2 *and then raise the result to the* 4^{th} *power before multiplying by* 0.05*. Multiplication is commutative; however, what Zion is describing is actually the associative property.*

   b. Evaluate the expression $[0.05 \times (3 \times 2)^4]^2$.
   $[0.05 \times 6^4]^2$
   $[0.05 \times 1{,}296]^2$
   $[64.8]^2$
   $4{,}199.04$

When using the order of operations to evaluate an expression with nested grouping symbols, apply the order of operations inside the inner grouping symbols first.

Consider providing additional time to discuss the absolute value component. Some students may have forgotten how absolute value works and need a minor refresher when reviewing.

Some students may struggle when working with nested grouping symbols. To help with this, have students mark the outer grouping symbols with green highlighter and underline everything between them. Then have them mark the inner grouping symbols with a yellow highlighter and draw a line above the symbols.

11.8.3 Collaboration: Determining Equivalent Expressions

Component Overview: *Students will work collaboratively with evaluating expressions. For this activity, students should work with a partner or small group to complete.*

11.8.3 Collaboration: Determining Equivalent Expressions

Answer the questions with your partner.

Tanisha says that when she evaluated an expression, the result was 7.25.

1. Use mental math to estimate each value. Then, circle which of the following expressions could have been Tanisha's expression.

$(6-0.5)^2 \times 5$ $\quad$ $6-0.5^2 \times 5$

$6+(-0.5)^2 \times 5$ $\quad$ $5 \times 6-0.5^2$

2. Find the values of the expressions to verify your choice.

Expression	Value
$(6-0.5)^2 \times 5$	151.25
$6-0.5^2 \times 5$	4.75
$6+(-0.5)^2 \times 5$	7.25
$5 \times 6-0.5^2$	29.75

Consider asking students which expressions in question 1 are equivalent. This may give some insight on how much the student has already learned.

Consider having students turn their calculators over for question 1 and then to use their calculators for question 2. Calculators such as Desmos can determine the value if the student inputs the expression as written. Consider providing students with four-function calculators, where the student will have to use their knowledge to evaluate while the calculator can be a support to learning.

11.8.4 Exploration: Grouping Symbols

Component Overview: *Students will work collaboratively to evaluate expressions involving grouping symbols. This activity is designed for students to complete with a partner.*

11.8.4 Exploration: Grouping Symbols

1. Without working the problems, discuss with your partner which expression in the table you expect to have the largest value.
Answer will vary. Discussion should be around law of exponents and order of operations to result in the largest value.

Expression A	Expression B
$10 - (0.75 \times 12) + 4^2$ $10 - 9 + 4^2$ $10 - 9 + 16$ 17	$(10 - 0.75) \times (12 + 4^2)$ $(9.25) \times (12+16)$ 9.25×28 259
Expression C	**Expression D**
$10 - 0.75 \times 12 + 4^2$ $10 - 0.75 \times 12 + 16$ $10 - 9 + 16$ 17	$10 - 0.75 \times (12 + 4)^2$ $10 - 0.75 \times (16)^2$ $10 - 0.75 \times 256$ $10 - 192$ -182

2. Put an asterisk (*) next to the expression you expect will have the largest value. Write your explanation here.

Answer will vary. Explanation will be based on the laws of exponents or order of operations used.

3. Evaluate each of the expressions in the table with your partner.

Prior to releasing this activity to students, consider reviewing the narrative with the students. Many students will jump straight to the expressions and begin evaluating and miss the key point of them making a guess as to which one they think will have the largest value. You may also want to consider having the class vote and writing the total number of votes per expression in a central area to refer back to when reviewing the answers.

11.8.4 Exploration: Grouping Symbols (continued)

4. With another set of partners, share if your guess was or was not correct. Discuss how the order of operations affected the value of your guesses. Summarize your discussion.
   Answers will vary. The discussion should be about order of operations causing expression D to be the largest value.

5. For the two expressions in the table that resulted in the same value, explain why they have the same value.
   Where the grouping symbol was placed did not have any effect on the order of operations. Multiplication would have been completed second in both expressions after evaluating the exponent.

?

What was the impact of the parentheses on the four expressions in this activity?

11.8.5 Your Turn!

Component Overview: *Students will practice evaluating expressions with multiple steps using the order of operations. For this activity, students should work independently to practice what they've learned in this lesson.*

11.8.5 Your Turn!

1. Evaluate each expression without using a calculator.

   a. $[(10-15)\times 2]^3+50$
   $[-5\times 2]^3+50$
   $(-10)^3+50$
   $-1000+50$
   -950

   b. $(2^4-3\times 6)^2$
   $(16-3\times 6)^2$
   $(16-18)^2$
   $(-2)^2$
   4

   c. $|18+(-2)\times 14|^5$
   $|18+(-28)|^5$
   $|-10|^5$
   10^5
   $100{,}000$

   d. $[0.8\times(1.2+2.8)^2]^2$
   $[0.8\times 4^2]^2$
   $[0.8\times 16]^2$
   12.8^2
   163.84

   e. $[(0.8\times 10)^2+4.5]\div 0.5$
   $[8^2+4.5]\div 0.5$
   $[64+4.5]\div 0.5$
   $68.5\div 0.5$
   137

   f. $(10\times 0.4)^2+\big(28.4\div(-4)\big)$
   $4^2+\big(28.4\div(-4)\big)$
   $4^2+(-7.1)$
   $16+(-7.1)$
   8.9

Watch for how students perform on question 1a during this activity. If they are performing poorly, consider pulling the students into small groups to work through one of the other questions and then releasing them to do the rest on their own.

11.8.6 Wrap-Up: Reflection

Component Overview: *This activity should be done independently to provide specific student-level data.*

11.8.6 Wrap-Up: Reflection

1. Complete each statement based on your learning today.

   - A barrier to my learning was ...
     Answers will vary.
   - I will try to overcome this by ...
     Answers will vary.
   - Today I am still unsure about ...
     Answers will vary.
   - To fill in this gap I intend to ...
     Answers will vary.

If students are able to get the correct answer but not identify the mistake, they may not fully comprehend what is occurring incorrectly in someone else's work.
If students are not able to locate a mistake in the problem, it will highlight a specific misconception that they have.
If students identify the mistakes correctly but the response is not correct, they may have made a calculation error.